

Table [climate-associated cluster candidates, full]. Candidate climate-associated LD clusters (primary analysis: Åland-excluded, LD-pruned Ω), with the eMLG gate. **Bold** = restricted set (eMLG BF(dB) ≥ 15).

Cluster	Chr	Position (Mb)	Size	Axis	Sig.	% sig.	eMLG BF(dB)	eMLG β	XtX	DI%
F63637	Chr24	1.26–1.82	272	PC1	205	75.4	38.4	+0.094	37	58.8
F63772	Chr24	1.76–1.82	83	PC2	60	72.3	36.5	+0.122	26	0.0
F17762	Chr5	3.04–3.34	11	PC2	7	63.6	30.7	+0.123	26	27.3
F41962	Chr13	5.84–6.18	50	PC2	7	14.0	21.8	+0.095	26	0.0
F64355	Chr24	1.93–5.39	35	PC1	7	20.0	21.2	+0.051	23	5.7
F6554	Chr2	10.04–10.09	24	PC1	12	50.0	19.7	+0.067	20	0.0
F63812	Chr24	2.03–2.31	23	PC1	10	43.5	18.9	+0.057	23	21.7
F27646	Chr8	6.24–6.80	486	PC1	108	22.2	14.9	+0.058	23	46.3
F67035	Chr26	1.55–4.72	397	PC2	19	4.8	14.4	+0.076	23	73.0
F49451	Chr16	0.43–5.91	522	PC1	69	13.2	13.7	+0.055	21	0.0
F17747	Chr5	2.82–4.27	690	PC1+2	24	3.5	11.5	+0.088	23	13.8
F11460	Chr3	6.19–8.49	1143	PC2	25	2.2	10.5	+0.079	21	0.6
F66997	Chr26	1.44–4.71	3042	PC2	37	1.2	10.1	+0.082	23	17.3
F48626	Chr15	8.19–8.21	19	PC1	5	26.3	8.2	+0.050	25	0.0
F14815	Chr4	6.45–11.04	99	PC2	19	19.2	7.6	+0.078	30	28.3
F27668	Chr8	0.96–6.91	400	PC2	12	3.0	7.2	+0.061	24	0.0
F64102	Chr24	3.08–4.36	28	PC1+2	7	25.0	5.7	+0.074	26	3.6
F67238	Chr26	6.56–7.48	42	PC2	7	16.7	5.0	+0.062	21	11.9
F51523	Chr17	0.05–4.77	3183	PC2	24	0.8	4.3	+0.058	19	0.0
F57265	Chr20	0.44–4.39	87	PC2	5	5.7	3.7	+0.063	21	0.0
F40144	Chr12	6.27–10.42	22	PC1	5	22.7	1.6	+0.039	20	4.5
F13363	Chr3	13.41–15.67	131	PC2	6	4.6	0.7	+0.048	23	0.8
F67071	Chr26	1.51–5.92	379	PC1	13	3.4	-2.5	+0.030	23	19.5
F51449	Chr17	5.11–5.74	893	PC1	6	0.7	-4.0	+0.028	20	5.0
F64194	Chr24	1.94–5.12	182	PC1	14	7.7	-6.4	+0.024	26	7.7
F68272	Chr27	0.05–3.21	496	PC2	8	1.6	-10.9	+0.018	19	12.1
F24440	Chr6	10.12–10.18	82	PC2	5	6.1	-11.2	+0.018	23	32.9
F67028	Chr26	1.82–6.75	1236	PC2	5	0.4	-11.9	+0.016	22	40.9
F67081	Chr26	6.29–7.03	449	PC2	5	1.1	-12.6	+0.014	22	51.0
F65755	Chr25	12.70–12.72	20	PC2	6	30.0	-13.4	+0.012	21	5.0
F67249	Chr26	7.36–7.48	71	PC2	26	36.6	-14.1	+0.011	22	62.0
F54612	Chr18	3.03–3.14	118	PC2	8	6.8	-14.5	+0.009	22	41.5
F7206	Chr2	9.72–10.10	64	PC1	28	43.8	-15.6	+0.007	26	1.6
F7494	Chr2	9.94–10.20	71	PC1	22	31.0	-16.4	+0.006	23	0.0
F6596	Chr2	10.00–10.10	14	PC1	5	35.7	-17.2	+0.005	23	0.0

Ordered by the cluster-level eMLG association on the axis the cluster entered on. Size = member markers; Axis = climate axis of entry (member-count); Sig. = member markers with BF(dB) ≥ 15 ; % sig. = Sig./Size; eMLG BF(dB) and β = the cluster's consensus-genotype association from a BayPass run on the eMLGs (primary config); XtX = mean among-population differentiation (a prerequisite for association, uniformly elevated); DI% = proportion of members with DI > -25 . **Bold** clusters pass the eMLG gate (BF(dB) ≥ 15 ; 7 of 35) and constitute the restricted candidate set; the rule separates them from clusters passing on member count alone. Genome-wide median cluster size = 9 markers.